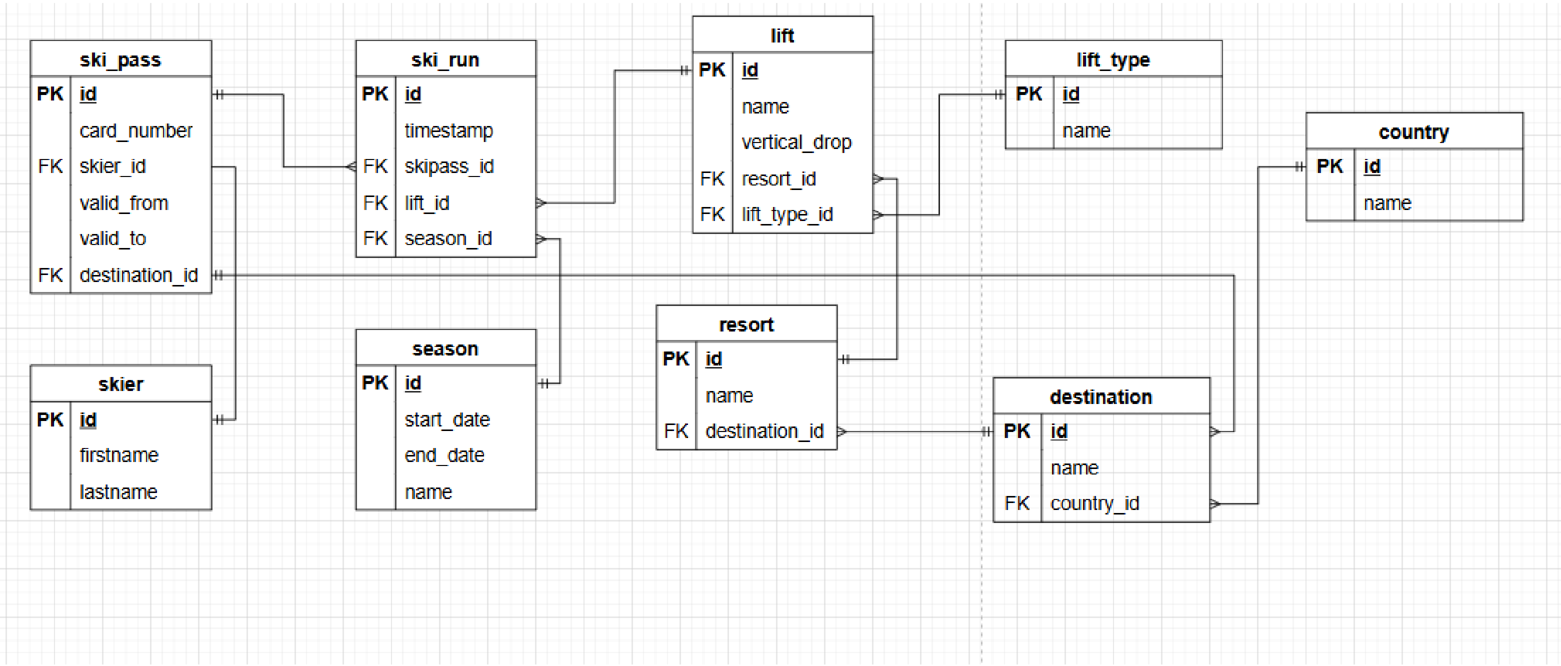
PROJEKTUPPGIFT



DELUPPGIFT 2

1. SELECT * FROM lift WHERE resort_id = 3

2. SELECT * FROM resort WHERE destination_id = 3

3. https://stackoverflow.com/questions/59093008/sql-query-to-get-the-data-between-two-numbers
   https://stackoverflow.com/questions/1233506/how-to-get-current-datetime-in-sql
   SELECT * FROM ski_pass
   WHERE id = 8 AND CURRENT_DATE BETWEEN valid_from AND valid_to

4. SELECT * FROM season
   WHERE CURRENT_DATE BETWEEN start_date AND end_date

5. SELECT firstname, lastname, username, "timestamp", lift.name FROM skier
   JOIN ski_pass ON ski_pass.skier_id = ski_pass.id
   JOIN ski_run ON ski_run.ski_pass_id = ski_pass.id
   JOIN lift on ski_run.lift_id = lift.id
   WHERE skier.id = 1 AND DATE_PART('day', "timestamp") = 25
   AND DATE_PART('month', "timestamp") = 3
   AND DATE_PART('year', "timestamp") = 2026

6. https://stackoverflow.com/questions/1521605/selecting-count-with-distinct
   SELECT firstname, lastname, COUNT(distinct ski_run.id) as runs,

```
COUNT(distinct lift.id) as lifts FROM skier
JOIN ski_pass ON ski_pass.skier_id = skier.id
JOIN ski_run ON ski_run.ski_pass_id = ski_pass.id
JOIN lift ON ski_run.lift_id = lift.id
WHERE skier.id = 1 AND DATE_PART('day', "timestamp") = 25
AND DATE_PART('month', "timestamp") = 3
AND DATE_PART('year', "timestamp") = 2026
GROUP BY skier.id
```

7. https://stackoverflow.com/questions/18152772/count-number-of-distinct-days-in-query
   https://hightouch.com/sql-dictionary/sql-sum

```
SELECT username, season.name, COUNT(distinct ski_run.id) as runs,
COUNT(distinct lift.id) as lifts,
COUNT(distinct DATE_PART('day', "timestamp")) as days,
SUM(lift.vertical_drop) as total_drop FROM season
JOIN ski_run ON ski_run.season_id = season.id
JOIN lift ON ski_run.lift_id = lift.id
JOIN ski_pass ON ski_run.ski_pass_id = ski_pass.id
JOIN skier ON ski_pass.skier_id = skier.id
WHERE season.id = 2 AND skier.id = 1
GROUP BY season.id, skier.id
```

8. https://stackoverflow.com/questions/75657502/extract-only-date-from-current-date-interval-data-type-in-postgres

```
SELECT username, season.name, COUNT(distinct ski_run.id) as runs,
COUNT(distinct lift.id) as lifts,
COUNT(distinct DATE_PART('day', "timestamp")) as days,
SUM(lift.vertical_drop) as total_drop FROM season
JOIN ski_run ON ski_run.season_id = season.id
JOIN lift ON ski_run.lift_id = lift.id
JOIN ski_pass ON ski_run.ski_pass_id = ski_pass.id
JOIN skier ON ski_pass.skier_id = skier.id
WHERE "timestamp" BETWEEN (CURRENT_TIMESTAMP - interval '1
week')::date AND CURRENT_TIMESTAMP
AND skier.id = 1
GROUP BY season.id, skier.id
```

9.
```
SELECT username, SUM(vertical_drop) as total_drop_m FROM skier
JOIN ski_pass ON ski_pass.skier_id = skier.id
JOIN ski_run ON ski_run.ski_pass_id = ski_pass.id
JOIN lift ON ski_run.lift_id = lift.id
JOIN resort ON lift.resort_id = resort.id
JOIN destination ON resort.destination_id = destination.id
```

```
WHERE DATE_PART('day', "timestamp") = 25
AND DATE_PART('week', "timestamp") = 13
AND DATE_PART('month', "timestamp") = 3
AND DATE_PART('year', "timestamp") = 2026
AND destination.id = 3
GROUP BY skier.id
ORDER BY total_drop_m DESC
```

Kommentar: sen är det bara att ta bort day till exempel så får man för senaste veckan osv..
Ta bort destination så får man totalen.

Övriga SQL frågor

Constraint ski_pass:

https://stackoverflow.com/questions/25422573/how-to-restrict-the-length-of-integer-when-creating-a-table-in-sql-server

-kortnummer ska va 9 siffror

```
ALTER TABLE ski_pass

ADD CONSTRAINT chk_card_number_length

CHECK(card_number between 99999999 and 1000000000)
```

Kommentar: kom på att detta är dumt eftersom kortnummer nu inte kan börja på noll men hinner ej ändra i allt.

Constraint skier

```
ALTER TABLE skier

ADD CONSTRAINT chk_name_valid_chars

CHECK (firstname ~ '^[A-Za-zÅÄÖåäö]+$')

--

ALTER TABLE skier

ADD CONSTRAINT chk_name_valid_chars_

CHECK (lastname ~ '^[A-Za-zÅÄÖåäö]+$')

--

ALTER TABLE skier

ADD CONSTRAINT chk_name_valid_chars_username

CHECK (username ~ '^[A-Za-zÅÄÖåäö123456789]+$')
```

--

ALTER TABLE skier

ADD CONSTRAINT chk_email_valid_chars

CHECK (email ~ '^[A-Za-zÅÄÖåäö123456789@.]+$')

---tänker att man har en säsongsklass och sedan en lifetime - klass typ.

---Information om skidåkare, 1-4.

SELECT firstname, lastname, ski_pass.valid_to as

slutdatum, COUNT(ski_run.id) as total_season_runs, season.name

as season, COUNT(distinct DATE_PART('day', ski_run.timestamp)) as total_season_days

FROM skier

JOIN ski_pass ON ski_pass.skier_id = skier.id

JOIN ski_run ON ski_run.ski_pass_id = ski_pass.id

JOIN season ON ski_run.season_id = season.id

WHERE skier.id = 2 AND DATE_PART('year', season.end_date) = DATE_PART('year', CURRENT_DATE)

GROUP BY skier.id, ski_pass.id, season.id

EDIT: Denna hämtar (förnamn, efternamn, slutdatum, totalt antal åk på säsongen, säsong och totalt antal dagar på säsongen)

---Information om skidåkare, 5-6.

SELECT firstname, lastname, SUM(lift.vertical_drop) as

total_vertical_drop,

COUNT(distinct country.id) as total_countries from skier

JOIN ski_pass ON ski_pass.skier_id = skier.id

JOIN ski_run ON ski_run.ski_pass_id = ski_pass.id

JOIN lift ON ski_run.lift_id = lift.id

JOIN resort ON resort.id = lift.resort_id

JOIN destination ON resort.destination_id = destination.id

JOIN country ON destination.country_id = country.id

WHERE skier.id = 2

GROUP BY skier.id

EDIT: Denna hämtar (förnamn, efternamn, totalt antal fallmeter, antal länder)

KÖPA LFITKORT

Först måste man ju typ ange sina uppgifter tänker jag typ en SELECT först då, sedan en funktion med insert som tar emot en skier id. Då måste vi kanske ha att användarnamn är unikt så att man kan ange sitt användarnamn.

https://stackoverflow.com/questions/13146304/how-to-select-every-row-where-column-value-is-not-distinct

inser att alla användarnamn inte är unika och måste söka upp alla användarnamn som inte är unika, SQL fråga:

SELECT * FROM skier WHERE username IN (SELECT username FROM skier GROUP BY username HAVING COUNT(*)>1)

ORDER BY username ASC

Tydligen har jag massa dubletter så måste ta bort dom.

1 - https://www.datacamp.com/tutorial/sql-remove-duplicates?utm_cid=23340058068&utm_aid=192632749329&utm_campaign=230119_1-ps-dscia~dsa-tofu~sql_2-b2c_3-emea_4-prc_5-na_6-na_7-le_8-pdsh-go_9-nb-e_10-na_11-na&utm_loc=9062437-&utm_mtd=-c&utm_kw=&utm_source=google&utm_medium=paid_search&utm_content=ps-dscia~emea-en~dsa~tofu~tutorial~sql&gad_source=1&gad_campaignid=23340058068&gbraid=0AAAAADQ9WsEQNEMQKXS2Z3bijLcBrBmfH&gclid=CjwKCAjwyYPOBhBxEiwAgpT8P_0uk0-pfSqWqL-mWPGHp1KJxcYDCuokiWIKUqNVIyTY8MDhLJnyBxoCuoEQAvD_BwE

2 - https://www.w3schools.com/sql/sql_join_self.asp

3- https://stackoverflow.com/questions/6583916/delete-duplicate-rows-from-small-table

-kontrollera dubletter: SELECT * FROM skier WHERE username IN (SELECT username FROM skier GROUP BY username HAVING COUNT(*)>1)

ORDER BY username ASC

-ta bort dubletter: DELETE FROM skier A USING (SELECT username

FROM skier GROUP BY username HAVING COUNT(*)>1) B

WHERE A.username = B.username

Radera resort

Använde ”on delete cascade” för att rader ska försvinna i tabeller som använder främmande nycklar.

https://www.youtube.com/watch?v=vANfY96ccOY

SQL frågor:

- Ta bort lift:

ALTER TABLE ski_run

ADD CONSTRAINT ski_run_lift_id_fkey

FOREIGN KEY(lift_id) REFERENCES lift(id)

ON DELETE CASCADE

- Ta bort resort:

ALTER TABLE lift

ADD CONSTRAINT lift_resort_id_fkey

FOREIGN KEY(resort_id) REFERENCES resort(id)

ON DELETE CASCADE